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TP N°2.2 — Laboratory Report

Clustering with R (K-means and CAH)

FDTA — Fouille de Données, Techniques et Applications

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Objectives

This lab demonstrates the application of two unsupervised clustering methods in R: K-means partitioning and Hierarchical Agglomerative Clustering (CAH). Using the fromage dataset (29 cheese varieties, 9 nutritional attributes), we explore group discovery through centroid-based and linkage-based approaches. The goal is to identify natural groupings of cheeses based on their nutritional profiles and compare the output of both techniques.

A — Data Loading and Description

Step 1: Setting the Working Directory

```
setwd("C:/Users/Younes/Desktop/TP_FDDTIA/TP2.2")
```

```
> setwd("C:/Users/Younes/Desktop/TP_FDDTA/TP2.2")
```

setwd() changes R's current working directory to the folder containing the dataset. This eliminates the need for absolute file paths in subsequent read operations, making the script portable. Any file referenced by name alone will now be resolved from this directory.

Step 2: Loading the Dataset

```
fromage <- read.table(file="fromage.txt", header=T, row.names=1, sep="\t", dec=".")
```

```
> fromage <- read.table(file="fromage.txt",header=T,row.names=1,sep="\t",dec=".")
```

read.table() imports the tab-delimited text file into an R data frame named fromage. The header=T argument uses the first row as column names, row.names=1 assigns cheese names as row identifiers, and dec="." ensures decimal values are correctly parsed under the current locale.

Step 3: Dataset Preview (RStudio Viewer)

	calories	sodium	calcium	lipides	retinol	folates	proteines	cholesterol	magnesium
CarredelEst	314	353.5	72.6	26.3	51.6	30.3	21.0	70	20
Babybel	314	238.0	209.8	25.1	63.7	6.4	22.6	70	27
Beaufort	401	112.0	259.4	33.3	54.9	1.2	26.6	120	41
Bleu	342	336.0	211.1	28.9	37.1	27.5	20.2	90	27
Camembert	264	314.0	215.9	19.5	103.0	36.4	23.4	60	20
Cantal	367	256.0	264.0	28.8	48.8	5.7	23.0	90	30
Chabichou	344	192.0	87.2	27.9	90.1	36.3	19.5	80	36
Chaource	292	276.0	132.9	25.4	116.4	32.5	17.8	70	25
Cheddar	406	172.0	182.3	32.5	76.4	4.9	26.0	110	28
Comte	399	92.0	220.5	32.4	55.9	1.3	29.2	120	51

The RStudio data viewer confirms that all 29 cheese varieties were imported correctly with 9 nutritional columns. Numeric columns such as calories, sodium, calcium, and retinol display consistent decimal formatting. Row names (cheese labels) serve as meaningful identifiers throughout the clustering analysis.

Step 4: Displaying the First 6 Rows

```
print(head(fromage))
```

```
> print(head(fromage))
            calories sodium calcium lipides retinol folates proteines cholesterol magnesium
CarredelEst      314  353.5    72.6    26.3    51.6    30.3      21.0          70        20
Babybel          314  238.0   209.8    25.1    63.7     6.4      22.6          70        27
Beaufort         401  112.0   259.4    33.3    54.9     1.2      26.6         120        41
Bleu             342  336.0   211.1    28.9    37.1    27.5      20.2          90        27
Camembert        264  314.0   215.9    19.5   103.0    36.4      23.4          60        20
Cantal           367  256.0   264.0    28.8    48.8     5.7      23.0          90        30
```

head() outputs the first six records to the console, providing a quick structural check on the loaded data. Beaufort leads in calcium (259.4) and Cantal in calcium as well (264.0), illustrating the high within-group variance that motivates clustering. The consistent presence of all nine numeric attributes confirms no columns were silently dropped during import.

Step 5: Descriptive Statistics

```
print(summary(fromage))
```

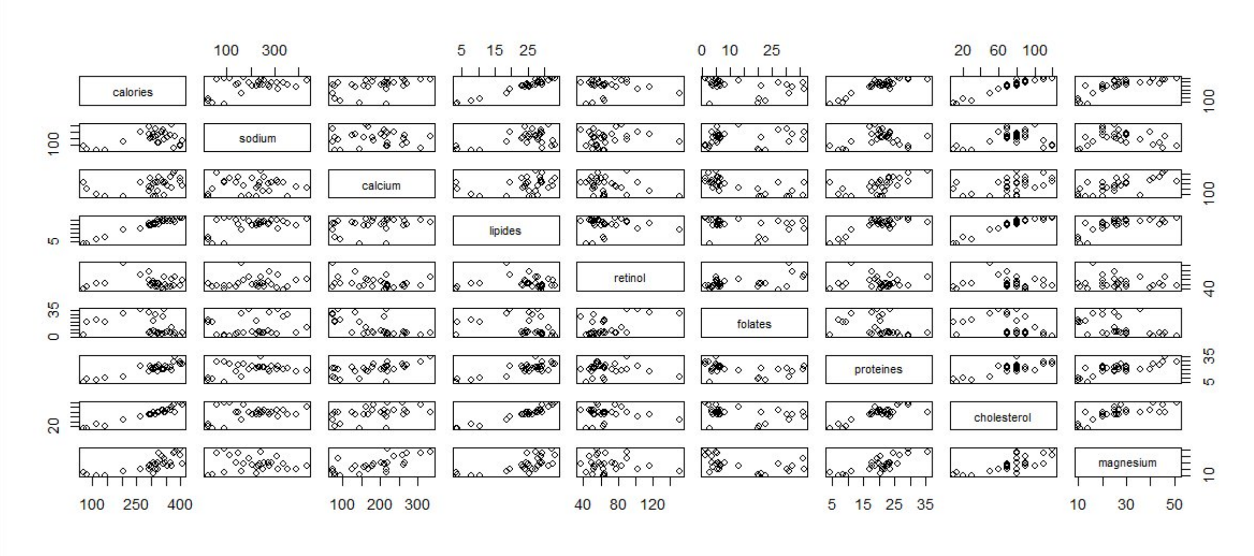
```
> print(summary(fromage))
    calories        sodium          calcium          lipides          retinol          folates     
 Min.   : 70   Min.   : 22.0   Min.   : 72.6   Min.   : 3.40   Min.   : 37.10   Min.   : 1.20  
 1st Qu.:292   1st Qu.:140.0   1st Qu.:132.9   1st Qu.:23.40   1st Qu.: 51.60   1st Qu.: 4.90  
 Median :321   Median :223.0   Median :202.3   Median :26.30   Median : 62.30   Median : 6.40  
 Mean   :300   Mean   :210.1   Mean   :185.7   Mean   :24.16   Mean   : 67.56   Mean   :13.01  
 3rd Qu.:355   3rd Qu.:276.0   3rd Qu.:220.5   3rd Qu.:29.10   3rd Qu.: 76.40   3rd Qu.:21.10  
 Max.   :406   Max.   :432.0   Max.   :334.6   Max.   :33.30   Max.   :150.50   Max.   :36.40  
   proteines      cholesterol       magnesium    
 Min.   : 4.10   Min.   : 10.00   Min.   :10.00  
 1st Qu.:17.80   1st Qu.: 70.00   1st Qu.:20.00  
 Median :21.00   Median : 80.00   Median :26.00  
 Mean   :20.17   Mean   : 74.59   Mean   :26.97  
 3rd Qu.:23.40   3rd Qu.: 90.00   3rd Qu.:30.00  
 Max.   :35.70   Max.   :120.00   Max.   :51.00  
```

summary() provides per-variable statistics including minimum, maximum, mean, and quartile values. Calories range from 70 to 406, while retinol spans 37.1 to 150.5, highlighting the disparity in scales across attributes. This scale heterogeneity is important to note: K-means uses Euclidean distance and may be sensitive to unnormalized variables.

Step 6: Pairwise Scatter Plot Matrix

```
pairs(fromage)
```

```
> pairs(fromage)
```



pairs() generates a scatter plot matrix showing all 36 variable combinations simultaneously. Visible clustering patterns in the lipides-calories and calcium-proteines sub-plots suggest natural groupings exist in the data. The matrix also reveals several near-linear relationships (e.g., calories vs. lipides), which will influence cluster separation.

B — Clustering by K-means

Step 7: Running K-means Clustering (k=4)

```
groupes.kmeans <- kmeans(fromage, centers=4, nstart=5)
```

```
> groupes.kmeans <- kmeans(fromage,centers=4,nstart=5)
```

kmeans() partitions the 29 cheeses into 4 clusters by iteratively minimising within-cluster sum of squares. Setting nstart=5 runs the algorithm from 5 different random initialisations and retains the best result, reducing sensitivity to initial centroid placement. The resulting object groupes.kmeans stores cluster assignments, centroids, and variance decomposition metrics.

Step 8: K-means Object Structure (Environment View)

Name	Type	Value
groupes.kmeans	list [9] (S3: kmeans)	List of length 9
cluster	integer [29]	4 1 3 1 1 1 ...
centers	double [4 x 9]	329.80 101.75 363.88 297.86 306.60 44.75 146.12 239.79 206.82 133.75 257.03 1...
totss	double [1]	763883.3
withinss	double [4]	63933 18690 59453 52672
tot.withinss	double [1]	194748.1
betweenss	double [1]	569135.2
size	integer [4]	10 4 8 7
iter	integer [1]	2
ifault	integer [1]	0

The RStudio environment panel shows that groupes.kmeans is a list of 9 components. Key fields include cluster (integer vector of 29 assignments), centers (4×9 centroid matrix), withinss (per-cluster SS), and betweenss/totss (global fit). The between-to-total SS ratio serves as a primary quality indicator for the partitioning.

Step 9: K-means Results — Cluster Assignments and Centroids

```
print(groupes.kmeans)
```

```
> print(groupes.kmeans)
K-means clustering with 4 clusters of sizes 10, 8, 4, 7

Cluster means:
  calories   sodium  calcium  lipides  retinol  folates proteines cholesterol magnesium
1 329.8000 306.6000 206.8200 26.87000 64.24000 11.91000  21.08000    82.00000  27.40000
2 363.8750 146.1250 257.0250 29.05000 63.60000  3.86250  26.56250    96.25000  38.87500
3 101.7500  44.7500 133.7500  6.27500 55.15000 16.47500   7.20000    18.25000  11.25000
4 297.8571 239.7857 103.8429 24.91429 83.92857 23.05714  18.97143    71.42857  21.71429

Clustering vector:
       CarredelEst            Babybel            Beaufort               Bleu          Camembert
                 4                  1                   2                  1                  1
            Cantal          Chabichou            Chaource            Cheddar              Comte
                 1                  4                   4                  2                  2
       Coulomniers               Edam            Emmental Fr.chevrepatemolle       Fr.fondu.45
                 4                  2                   2                  4                  1
     Fr.frais20nat.     Fr.frais40nat.           Maroilles            Morbier           Parmesan
                 3                  3                   1                  1                  2
      Petitsuisse40        PontlEveque            Pyrenees          Reblochon         Rocquefort
                 3                  4                   1                  1                  1
        SaintPaulin               Tome            Vacherin Yaourtlaitent.nat.
                 2                  4                   2                  3

Within cluster sum of squares by cluster:
[1] 63933.23 59452.98 18690.22 52671.70
 (between_SS / total_SS =  74.5 %)

Available components:

[1] "cluster"      "centers"      "totss"        "withinss"     "tot.withinss" "betweenss"
[7] "size"         "iter"         "ifault"
```

The printed output reveals four clusters of sizes 10, 8, 4, and 7. Cluster 3 groups low-calorie fresh cheeses (Fr.frais20nat., Petitsuisse40, Yaourtlaitent.nat.) with markedly lower lipide and cholesterol centroids. The between_SS / total_SS ratio of 74.5% confirms that the four-cluster solution captures a substantial portion of the total variance in the dataset.

C — Hierarchical Agglomerative Clustering (CAH)

Step 10: Computing the Distance Matrix

```
d.fromage <- dist(fromage)
```

```
> d.fromage <- dist(fromage)
```

Values	
d.fromage	'dist' num [1:406] 181 324 145 166 224…

dist() computes the Euclidean distance between all 29×28/2 = 406 unique cheese pairs. The resulting dist object d.fromage encodes the full pairwise dissimilarity matrix in a compact lower-triangular format. This distance matrix forms the sole input to the hierarchical clustering step, making the choice of distance metric a critical modelling decision.

Step 11: Building the Hierarchical Clustering (Complete Linkage)

```
cah_clink <- hclust(d.fromage, method="complete")
```

```
> cah_clink <- hclust(d.fromage,method="complete")
```

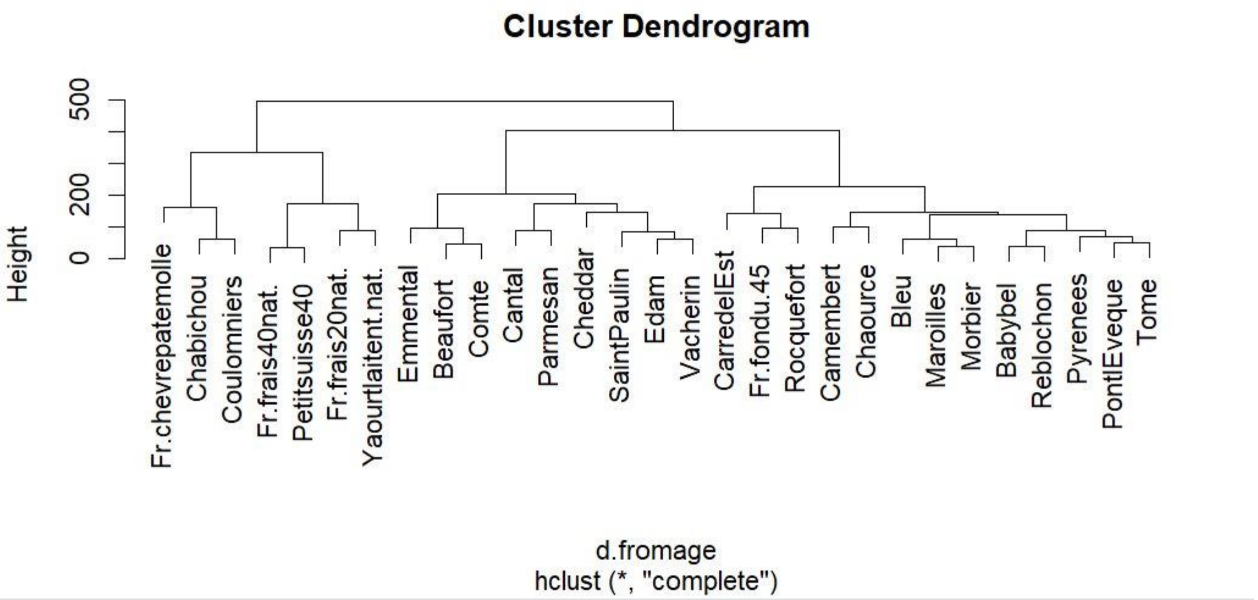
Name	Type	Value
cah_clink	list [7] (S3: hclust)	List of length 7
merge	integer [28 x 2]	-17 -18 -2 -3 -22 -12 -21 -19 -24 -10 -27 -28 ...
height	double [28]	33.6 37.5 38.0 45.0 49.7 59.8 ...
order	integer [29]	14 7 11 17 21 16 ...
labels	character [29]	'CarredelEst' 'Babybel' 'Beaufort' 'Bleu' 'Camembert' 'Cantal' ...
method	character [1]	'complete'
call	language	hclust(d = d.fromage, method = "complete")
dist.method	character [1]	'euclidean'

hclust() performs agglomerative clustering using the complete-link criterion, where inter-cluster distance equals the maximum pairwise distance between members. Complete linkage produces compact, roughly spherical clusters and is less sensitive to outliers than single linkage. The resulting object cah_clink stores the merge sequence, heights, and Euclidean distance method used.

Step 12: Plotting the Dendrogram

```
plot(cah_clink)
```

```
> plot(cah_clink)
```

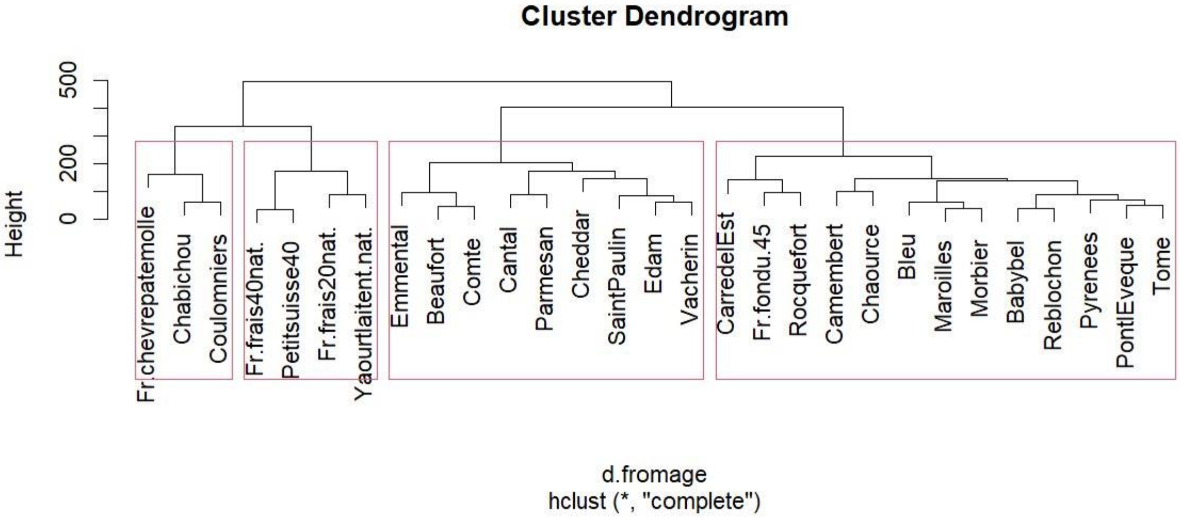


plot() renders the full hierarchical dendrogram with cheese names at the leaves and fusion heights on the y-axis. The large height gap between the penultimate and final merges (around height 300-500) visually suggests that cutting at k=4 is a natural partition. The branching structure clearly separates the low-calorie fresh cheeses (left subtree) from the richer aged varieties (right subtree).

Step 13: Dendrogram with 4-Cluster Rectangles

```
rect.hclust(cah_clink, k=4)
```

```
> rect.hclust(cah_clink,k=4)
```



rect.hclust() overlays coloured rectangles on the dendrogram to materialise the four chosen groups visually. The rectangles confirm two compact left-side clusters (soft/fresh cheeses) and two larger right-side clusters (hard and semi-hard varieties). This visualisation makes it straightforward to communicate cluster membership to a non-technical audience.

Step 14: Cutting the Dendrogram into 4 Groups

```
groupes.cah <- cutree(cah_clink, k=4)
```

```
> groupes.cah <- cutree(cah_clink,k=4)
```

groupes.cah	Named int [1:29] 1 1 2 1 1 2 3 1 2 2 ...

cutree() slices the dendrogram at the level that produces exactly 4 clusters, returning a named integer vector of group assignments for all 29 cheeses. The RStudio environment confirms the resulting groupes.cah variable as a Named int [1:29] with values ranging from 1 to 4. This object is now ready for further profiling, comparison with K-means labels, or export.

Step 15: Sorted Group Membership Listing

```
print(sort(groupes.cah))
```

```
> print(sort(groupes.cah))
       CarredelEst            Babybel               Bleu          Camembert           Chaource
                 1                  1                  1                  1                  1
        Fr.fondu.45          Maroilles            Morbier        PontlEveque           Pyrenees
                 1                  1                  1                  1                  1
         Reblochon         Rocquefort               Tome           Beaufort             Cantal
                 1                  1                  1                  2                  2
           Cheddar              Comte               Edam           Emmental           Parmesan
                 2                  2                  2                  2                  2
       SaintPaulin           Vacherin          Chabichou        Coulomniers Fr.chevrepatemolle
                 2                  2                  3                  3                  3
     Fr.frais20nat.     Fr.frais40nat.       Petitsuisse40 Yaourtlaitent.nat.
                 4                  4                  4                  4
```

sort() reorders the result by cluster label, producing a grouped listing of all 29 cheeses. Cluster 1 is the largest, containing mainstream ripened cheeses (Babybel, Bleu, Camembert, Rocquefort, etc.), while Cluster 4 holds only the lightest fresh dairy products. Comparing this CAH partition to the K-means result reveals broad agreement in separating fresh from aged varieties, though minor reassignments highlight the algorithmic differences between the two methods.

Conclusion

This laboratory demonstrated the application of two complementary clustering algorithms on the fromage nutritional dataset. K-means produced four well-separated groups (between_SS/total_SS = 74.5%) with clear centroid profiles distinguishing caloric, protein-rich aged cheeses from low-fat fresh varieties. The CAH complete-link dendrogram confirmed the same major divisions while revealing finer sub-structure through its hierarchical merge sequence. Both methods converged on a consistent four-cluster partition, reinforcing the validity of the discovered groups and demonstrating the complementary value of visual (dendrogram) and quantitative (K-means metrics) clustering diagnostics.